Hyo Seo Kim

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RESEARCH INTERESTS

Simply training on more data is not always enough when pretrained models need to forget, merge, or continually adapt. My research focuses on **reliable model adaptation**. I study how relationships among alternative model updates can reveal what should be changed, and use this signal to design more reliable models.

PUBLICATIONS

HyperFix: Combinatorial Nonlinear Correction for Task Vector Merging
Hyo Seo Kim, *Ren Wang*
arXiv preprint, 2026
HyperFix learns a shared correction function for model merging across the combinatorial space of task subsets.

MoCo-EA: Exploiting Adversarial Mode Connectivity for Efficient Evolutionary Attacks
Hyo Seo Kim, *Gang Luo, Can Chen, Binghui Wang, Yue Duan, Ren Wang*
ICML 2026 & ICLR 2026 Workshop on Principled Design for Trustworthy AI (Oral)
MoCo-EA reveals that adversarial examples lie on connected manifolds rather than isolated points.

NegMerge: Sign-Consensual Weight Merging for Machine Unlearning
Hyo Seo Kim, *Dongyoon Han*, Junsuk Choe* (* corresponding author)*
ICML 2025 & NeurIPS 2024 Workshop on Adaptive Foundation Models
NegMerge introduces model merging into machine unlearning by combining sign-consistent task vectors.

Weakly-supervised Incremental learning for Semantic segmentation with Class Hierarchy
Hyo Seo Kim, *Junsuk Choe*
Pattern Recognition Letters 2024
WISH leverages class hierarchy for weakly supervised class-incremental semantic segmentation.

Curriculum Learning with Class-label Composition for Weakly Supervised Semantic Segmentation
Dongjun Hwang, Hyo Seo Kim, *Doyeol Baek, Hyunbin Kim, Inhye Kye, Junsuk Choe*
Pattern Recognition Letters 2025
Introduces curriculum learning based on the length and frequency of class-label compositions.

EDUCATION

Illinois Institute of Technology, Chicago, IL — **Aug. 2025 – May. 2029 (Expected)**
Ph.D. in Computer Engineering — *Advisor: Prof. Ren Wang*

Sogang University, Seoul, Korea — **Aug. 2022 – Aug. 2025**
M.S. in Computer Science and Engineering — *Advisor: Prof. Junsuk Choe*

Philipps University, Marburg, Germany — **Feb. 2021 – Aug. 2021**
Exchange Student in Economics

Sogang University, Seoul, Korea — **Feb. 2018 – Aug. 2022**
B.A. in Economics, and B.S. in Computer Science and Engineering — *Graduated with Cum Laude*

WORK EXPERIENCE

Michigan State University, East Lansing, MI **Oct. 2024 – Apr. 2025**
Visiting Scholar *Host: Prof. Sijia Liu*
– Conducted research on machine unlearning for large language models and presented the work at the 9th Annual International Scholar Showcase at Michigan State University.

NAVER Cloud, Seongnam, Korea **Jul. 2024 – Oct. 2024**
Research Intern *Mentor: Dr. Dongyoon Han*
– Conducted machine unlearning research on vision-language and image classification models, resulting in NegMerge (ICML 2025).

Tübingen AI Center, Tübingen, Germany **Jul. 2023 – Aug. 2023**
Visiting Scholar *Host: Dr. Seong Joon Oh*
– Engaged in research discussions and paper study sessions with researchers at the Tübingen AI Center.

SK C&C, Seoul, Korea **Jun. 2022 – Aug. 2022**
Web Developer Intern *Received Full-Time Return Offer*
– Migrated a sales system from Nexacro to WebSquare5, developing 12+ screens and resolving 10+ defects.

Sogang BaSE Lab, Seoul, Korea **Aug. 2021 – Aug. 2022**
Undergraduate Intern *Advisor: Prof. Sooyong Park*
– Developed a cryptocurrency payment approval system, resulting in a publication (Korea Computer Congress).

LG CNS, Seoul, Korea **May. 2022 – Jun. 2022**
ERP Developer Intern *Received Full-Time Return Offer*
– Developed a system to track the approval status of expense vouchers for specific recipients.

LIKELION, Seoul, Korea **Dec. 2020 – Mar. 2021**
QA Intern
– Performed QA for web services and developed supplementary materials for programming courses.

ACADEMIC SERVICE

– **Conference Volunteer:** IEEE MASS 2025
– **Conference Reviewer:** CPAL 2025, ICLR 2026 Workshop on Principled Design for Trustworthy AI
– **Journal Reviewer:** Pattern Recognition Letters (Elsevier) 2025, IEEE Transactions on Neural Networks and Learning Systems 2026

TEACHING EXPERIENCE

Electrical and Computer Engineering, Illinois Institute of Technology **Chicago, IL**
Teaching Assistant
– ECE 572: Secure Machine Learning Design and Applications (Fall 2026).

Computer Science and Engineering, Sogang University **Seoul, Korea**
Teaching Assistant, Taught in English
– CSEG321: Introduction to Natural Language Processing (Fall 2023).
– CSE3080: Data Structures (Spring 2023).
– CSE4120: Fundamentals of Compiler Construction (Fall 2022).

Sogang Institute of Whole Person Education **Seoul, Korea**
Undergraduate Mentor
– COR1010: Introduction to AI Programming - Python (Spring 2022).
– COR1009: Computational Thinking - Python (Spring 2020, Fall 2020).

Sogang Graduate School of AI · SW **Seoul, Korea**
Teaching Assistant
– Data Science & AI/Software Engineering/IoT/Information Security/Blockchain (Spring 2023, Fall 2023, Spring 2024).